

Singularities

Basic skills

Q1. a) $\begin{vmatrix} 8 & -4 \\ 3 & 6 \end{vmatrix} = 48 + 12 = 60$

b) $\begin{vmatrix} 3 & -2 & -5 \\ 0 & 2 & 3 \\ 4 & 3 & 2 \end{vmatrix} = 3 \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} + 2 \begin{vmatrix} 0 & 3 \\ 4 & 2 \end{vmatrix} - 5 \begin{vmatrix} 0 & 2 \\ 4 & 3 \end{vmatrix}$
 $= 3(4 - 9) + 2(0 - 12) - 5(0 - 8)$
 $= -15 - 24 + 40 = 1$

Q2. a) $\begin{pmatrix} 4 & 1 \\ 8 & 2 \end{pmatrix}$ has determinant $8 - 8 = 0$
so singular

b) $\begin{pmatrix} 2 & -15 \\ -1 & 5 \end{pmatrix}$ has determinant $10 - 15 = -5$
so not singular

c) $\begin{pmatrix} 4 & 4 & -2 \\ 5 & -6 & 1 \\ 3 & -2 & 0 \end{pmatrix}$ has determinant
 $4 \begin{vmatrix} -6 & 1 \\ -2 & 0 \end{vmatrix} - 4 \begin{vmatrix} 5 & 1 \\ 3 & 0 \end{vmatrix} - 2 \begin{vmatrix} 5 & 1 \\ 3 & 0 \end{vmatrix}$
 $= 4 \times 2 - 4 \times -3 - 2 \times -3 = 8 + 12 + 6 = 26$
so not singular

Q3. If $ad=bc$ it means that the top line and bottom line are in proportions as $\frac{a}{b} = \frac{c}{d}$

e.g. $\begin{pmatrix} 2 & -1 \\ -4 & 2 \end{pmatrix}$ $(2, -1)$ and $(-4, 2)$ are multiples / Parallel vectors

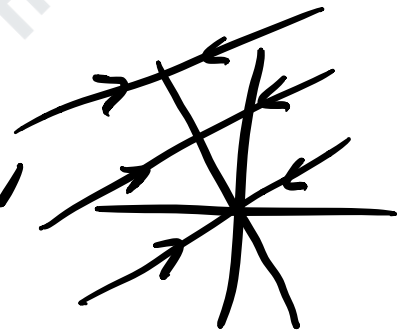
Competency skills

Q4. $M = \begin{pmatrix} 2 & -1 \\ -4 & 2 \end{pmatrix}$

a) $M \begin{pmatrix} x \\ x \end{pmatrix} = \begin{pmatrix} x \\ -2x \end{pmatrix}$ so $y = -2x$

b) $M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x - y \\ -4x + 2y \end{pmatrix}$

so $x' = 2x - y$
 $y' = -4x + 2y$ $y' = -2x'$



so All points go to $y = -2x$

c) more than a single point maps to a single point so unable to undo

like ↗ one to many

$$d) M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ -10 \end{pmatrix}$$

$$\Rightarrow \begin{matrix} 2x - y = 5 \\ -4x + 2y = -10 \end{matrix} \quad \text{so line } y = 2x - 5$$

All of $y = 2x - 5$ maps to $(5, -10)$

$$Q5. \begin{vmatrix} 4 & k & -2k \\ 5 & -6 & 1 \\ 3 & k+1 & 0 \end{vmatrix} = 4 \begin{vmatrix} -6 & 1 \\ k+1 & 0 \end{vmatrix} - k \begin{vmatrix} 5 & 1 \\ 3 & 0 \end{vmatrix} - 2k \begin{vmatrix} 5 & -6 \\ 3 & k+1 \end{vmatrix}$$

$$= 4(0 - (k+1)) - k(0 - 3) - 2k(5(k+1) + 18)$$

$$= -4k - 4 + 3k - 10k(k+1) - 36k$$

$$= -10k^2 - 47k - 4$$

$$\text{so } -10k^2 - 47k - 4 = 0$$

$$\text{so } 10k^2 + 47k + 4 = 0 \quad k = \frac{-47 \pm \sqrt{47^2 - 4 \times 10}}{2 \times 10}$$

$$k = -0.0867 \text{ or } -4.613$$

Q6. $M = \begin{pmatrix} 2 & 3 \\ 6 & 9 \end{pmatrix}$

For line of singularity:

$$M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x+3y \\ 6x+9y \end{pmatrix} \text{ so } \begin{matrix} x' = 2x+3y \\ y' = 6x+9y \end{matrix}$$

so $y' = 3x'$ so line is $y = 3x$

For Invariant points:

$$M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{matrix} x = 2x+3y \\ y = 6x+9y \end{matrix}$$

$$\Rightarrow x+3y=0, \quad 6x+8y=0$$

$$\text{so } y = -\frac{1}{3}x, \quad y = -\frac{3}{4}x \Rightarrow x=y=0$$

For Invariant lines:

$$M \begin{pmatrix} x \\ mx+c \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix} \Rightarrow \begin{matrix} x' = 2x+3mx+c \\ y' = 6x+9mx+c \end{matrix}$$

want $y' = mx' + c$

so as $y' = 3x'$ only line is if $m=3$
and $c=0$

$$\text{so } y = 3x$$

line of singularity is
a line of invariance

Q7. $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ collapses 2D space to $(0,0)$
the OD 'point'

Q8. $M = \begin{pmatrix} 3 & -1 & 2 \\ -1 & 0 & 2 \\ 4 & -2 & 8 \end{pmatrix}$

a) $M \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$ $\begin{matrix} x' = 3x - y + 2z \\ y' = -x + 2z \\ z' = 4x - 2y + 8z \end{matrix}$

So $\begin{matrix} x' - y' = 4x - y \\ 4x' - z' = 8x - 2y \end{matrix}$

So $x' - y' = 4x' - z' \Rightarrow 3x' + y' - z' = 0$

So Plane $3x + y - z$ is Plane of singularity

b) $M \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ can't have a unique sol
if the a, b, c are off plane

e.g. $\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ has $3 \times 1 + 1 - 1 = 3 \neq 0$
so off plane

c) For infinite solutions need to map to plane

$$\text{So } 3a - b - c = 0$$

$$\text{e.g. } \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \text{ has solutions}$$

In fact need x, y, z s.t

$$1 = 3x - y + 2z$$

$$2 = -x + 2z$$

$$1 = 4x - 2y + 8z$$

$$\text{So } 4x - y = -1 \Rightarrow y = 4x + 1$$

$$\text{and } 9 = -2y + 16z \Rightarrow y = 8z - \frac{9}{2}$$

$$\text{So line } 4x + 1 = y = 8z - \frac{9}{2}$$

$$\text{maps to } \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$